
SUSTAINABILITY × ARTIFICIAL INTELLIGENCE

Sustainable AI

AI **for** sustainability — and the sustainability **of** AI

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We build the data behind sustainability for companies

30+

regulatory data engagements

B2B and B2C · food, consumer goods,
industrials, retail

200+

people in DACH

Impact Strategies · Ettlingen, 15 minutes
from here

1,900

Goforeans

digital engineering, design, data,
strategy

25

offices across Europe

Gofore Group, listed in Helsinki

Impact Strategies is the strategy consulting arm of Gofore Group. Five practice areas — regulatory & ESG, business transformation, circular economy, decarbonisation, and ESG × AI, which is today.

BEFORE WE START

Is AI net-positive or net-negative for sustainability?

NET-POSITIVE

NET-NEGATIVE

NO IDEA

SUSTAINABILITY × ARTIFICIAL INTELLIGENCE

01 **AI for sustainability**

What it does for a company's climate, compliance and product data.

PART 2

02 **The sustainability of AI**

What AI itself costs — in energy, in materials, and in who gets access.

PART 3

03 **Where the two meet**

The net-value test — and whether you can show the arithmetic.

THE WHOLE TALK

PART 1

The frame — why climate and AI arrived on the same agenda

01

THE RISK MAP

Ten risks, three editions — the top has not moved, and the one that did move is AI

#	Global risk — ten-year horizon (2026 order)	2024	2025	2026	2-year 2026	Category
1	Extreme weather events	1	1	1	4	Environmental
2	Biodiversity loss and ecosystem collapse	3	2	2	> 10	Environmental
3	Critical change to Earth systems	2	3	3	> 10	Environmental
4	Misinformation and disinformation	5	5	4	2	Technological
5	Adverse outcomes of AI technologies	6	6	5	30	Technological
6	Natural resource shortages	4	4	6	> 10	Environmental
7	Inequality	> 10	7	7	7	Societal
8	Cyber insecurity*	8	9	8	6	Technological
9	Societal polarization	9	8	9	3	Societal
10	Pollution	10	10	10	9	Environmental

An environmental risk has led the ten-year list in every edition since 2022. What moved is AI — up to #5 — and, on the short-term list, geopolitics to #1.

WEF Global Risks Reports 2024–2026, ranked by severity; perception survey of ~1,300 experts and leaders. *Named “cyber espionage and warfare” in the 2025 edition. “> 10” = outside the top 10.

The market already pays for this pair — and very few people have both halves

1

AI and big data

The fastest-growing skill to 2030.

Learnable — prompting in an afternoon, the data chain in a project.

2

Environmental stewardship

A top-10 skill in the same ranking.

Not learnable in an afternoon — this half needs a domain: a heat pump, a packaging line, a supply chain.

3

The pair

+78 million net new jobs by 2030, ≈34 million of them from the green transition.

Environmental engineers are among the 15 fastest-growing professions.

The scarce profile is not the data scientist and not the sustainability manager — it is the person who is credible in both rooms.

WEF Future of Jobs Report 2025 — employer survey, 1,000+ companies. Expectations, not outcomes.

THE COMMON DENOMINATOR

Risk, market and regulation arrive as one request — know your chain, trust your data, steer by it

1

Transparency across the chain

Where from, under what conditions, at what footprint? Everyone asks the same.

36 jurisdictions, over half of world GDP, adopt ISSB standards — Europe is one case, not the case.

2

Data as the foundation

You can only steer what you can measure. Primary data, one definition per object, a named owner.

The part no software buys for you.

3

Insight to steer and optimise

From reporting to decisions: energy, materials, sourcing, routing, risk.

Where the value sits — and where AI enters.

The regulation is the least interesting of the three drivers. The capability they all demand is the same — and building it is what Part 2 is about.

THE HARD TRUTH

Sustainability only happens **when it creates value.**

Not from obligation alone. Measures survive the budget round when they pay off — in avoided cost, in efficiency, or in revenue. Everything else is a pilot with an expiry date.

That is exactly why AI matters: it changes what pays off — and what it costs.

PART 2

The value — what is in it for a company

02

A company acts in exactly three lanes — and regulation lives inside the first, not beside it

1

De-risk

Compliance & market access. Avoid fines, pass audits, keep the listing.

Most mature — table stakes. Anchored in disclosure rules and customer scorecards.

2

Cost-out

Efficiencies & cost. Less manual work, one shared data model.

The entry point — fastest visible savings. Measured in analyst hours and cost per SKU.

3

Growth-in

Revenue. Verified claims, premium positioning, data as part of the product.

Least mature — and where you can differentiate. Measured in win rate and price premium.

A measure that pays in none of the three does not survive the budget round. Name the lane before you name the tool.

ROI = benefit ÷ invest. AI works on both sides: it lowers the invest and raises the benefit.

THE BUSINESS CASE

“What does it cost us? What do we stop paying for? What can we sell?” — a CFO asks one question per lane

1

“What does it cost us if we cannot answer?”

Lane 1 · measure audit findings, days to answer a customer request, share of supplier data complete. Tail risk: a lost listing — and EUDR fines of at least 4 % of EU turnover.

2

“What do we stop paying for?”

Lane 2 · measure analyst hours per report, cost per SKU, rework rate. Vendor benchmarks: –70 % accounting time, –80 % manual effort — indicative, not audited.

3

“What can we sell that we could not sell before?”

Lane 3 · measure win rate in tenders and realised price premium. Put a verified product carbon footprint on one offer and track it.

There is no sustainability-AI business case without a baseline. Four weeks of measuring beats a six-month pilot.

The problem is not missing data — it is missing structure

1

Fix the structure
one requirement traced to one owner · one definition per data object

2

Fix the data
primary instead of estimated · complete · auditable

3

Enable AI
it accelerates what is structured — and automates the gaps in what is not

4

Win in the three lanes
de-risk · cost-out · growth-in

The order most companies buy: tool first, then the data, then the structure — and the tool automates the gaps, faster and with more confidence than the data deserves.

AI does not fix bad data. It makes bad data faster.

Four kinds of “AI” do four different jobs — choose by data type, not by hype

1

Workflows

Deterministic, explainable, negligible energy.

Use when frequency is high and rules are clear.

2

Generative AI

Reads and writes text. Probabilistic — demand citations.

For text-heavy inputs. Highest energy per task.

3

Predictive ML

Forecasts, risk scores and anomalies.

Use when there is enough history to learn from.

4

Specialist software

Validated methods and audit trails.

Use when regulatory depth dominates.

Structured & frequent → workflows · unstructured text → generative · forecast or risk → predictive · regulatory depth → specialist software.

PART 3

The critique — what AI costs, who pays, and how to know it is worth it

03

GUESS BEFORE THE REVEAL

What share of the world's electricity do data centres use today?

< 1 %

1–2 %

5 %

10 % +

THE FOOTPRINT

About 1.5 % of world electricity today — about 3 % by 2030, more than all of Germany uses

≈ 1.5 %

of global electricity today

415 TWh in 2024, 485 TWh in 2025 — the second box was right

≈ 3 %

by 2030 — about 950 TWh

IEA base case: roughly a doubling in five years, +17 % in 2025 alone

≈ 500 TWh

all of Germany, per year

the world's data centres pass one industrial country before 2030

PER TASK

Per task, the machine is the frugal one — a prompt is seconds of television, and a person writing the same page emits hundreds of times more

0.24 Wh

one median text prompt (Gemini, 2025)

≈0.03 g CO₂e and five drops of water — under nine seconds of television

130–1,500×

less CO₂e per written page than a person

Tomlinson et al. 2024 — with the model's training included

≈80 %

of that AI figure was training, amortised

leave training out and the per-task gap widens further

THE NET BALANCE

On the IEA's own numbers the credit is larger than the bill — but it is a potential

THE BILL — WHAT AI CONSUMES

≈ 0.35 Gt

data-centre CO₂ in 2035

About 2 % of global electricity-sector emissions.
Still growing: ≈950 TWh by 2030.

vs

THE CREDIT — WHAT AI COULD AVOID

≈ 1.8 Gt

avoidable emissions in 2035

The IEA's ≈5 % of energy-related emissions, in absolute terms.
Roughly five times the bill — if deployed.

It arrives only if four things hold: the solutions are deployed · the saving is measured, not assumed · the rebound is contained · the grid can carry it, in the right place and hour.

Behind every prompt: one dominant supplier at every layer of the chain — and no country holds two

99 %

**raw materials —
China**

gallium and rare-earth processing; export controls since 2023

100 %

**lithography —
Netherlands**

ASML, the only maker of EUV machines

≈90 %

wafers — Taiwan

TSMC, advanced nodes — the frontier every AI chip depends on

79 %

**HBM memory —
Korea**

SK Hynix and Samsung — the high-bandwidth memory beside every GPU

≈92 %

GPUs — United States

Nvidia and AMD, the chips themselves

And at the end of the chain: 1.2 to 5 million tonnes of e-waste from generative AI by 2030 — hardware retired after three to five years. Circular strategies could cut that by up to 86 %.

Six billion people are online, one in six uses AI — and the divide is widening, not closing

2.2 bn

people still offline

ITU 2025 — six billion online, about three quarters of the world

23 %

internet use in low-income countries

against 94 % in high-income countries; 5G covers 4 % vs 84 %

1 in 6

people worldwide use generative AI

North 24.7 %, South 14.1 % — the gap widened in 2025

2 in 3

children without internet at home

1.3 billion children aged 3–17, UNICEF–ITU

A US\$20 subscription is a rounding error in this room and about a fifth of average income in a low-income country. Who gets the productivity gain decides whether AI narrows inequality or widens it.

Sustainable AI is an engineering problem — five levers, and two of them are thermal

- | | | |
|---|--|--|
| 1 | Right-size the model | Classifying a certificate does not need a frontier model. Smaller models also run on modest hardware and in more languages — the access lever, too. |
| 2 | Clean power, right place and hour | Shift workloads to regions and hours with surplus renewables. Owner: cloud providers and grid operators — the demand side of the grid. |
| 3 | Cool smarter, reuse the heat | Liquid and water-free cooling; waste heat upgraded by heat pumps for district heating. Owner: data-centre engineering, utilities, municipalities — your field. |
| 4 | Keep the hardware longer, then recover it | Three-to-five-year GPU cycles drive the e-waste; life extension and recovery could cut it by 16–86 %. Owner: whoever buys compute. |
| 5 | Net-value test per use case | Run AI only where the balance is positive — and be able to show it. Owner: every company deploying AI. The only lever a user controls directly. |

In Germany the waste heat is already a legal quota — 10 % reused from July 2026, 20 % by 2028. Somebody has to engineer that.

THE METHOD

Five questions per use case — and a use case that fails any of them does not get built

- | | | |
|---|-------------------------------|--|
| 1 | Name the lane | De-risk, cost-out or growth-in — and the single KPI that will move. If the answer is “all three”, it is a slide, not a use case. |
| 2 | Name the kind of AI | From the toolbox. If a frontier model is not needed, do not use one — that is a footprint decision, not only a cost one. |
| 3 | Measure the baseline | Four weeks, before any tool: hours per report, cost per SKU, days to answer a customer. |
| 4 | Estimate the footprint | Inference energy per task × expected volume. An order of magnitude is enough. Most companies have never looked. |
| 5 | Compare, decide, sign | Savings enabled minus footprint caused — approved by a named person who owns the KPI. |

One afternoon per use case — and it kills a bad one in week one instead of month nine.

THE SAME QUESTION

Is AI net-positive or net-negative for sustainability — and has your answer moved on any of the three questions?

NET-POSITIVE

NET-NEGATIVE

IT DEPENDS

LET'S TALK

Where is the net value, for whom — and can I show it?

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The 36-slide version with every number, source and worked example is the hand-out.